

A GENERALIZED SELECTION GOAL FOR A RANKING PROBLEM

W.K. Chiu*

University of Bath

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SUMMARY

This paper proposes a generalised goal for the ranking problem of selecting the best from among several normal populations, using the $\{\delta^*, P^*\}$ -approach. The main result states that the probability of selecting a population having a mean within δ^* from the largest mean is at least P^* , if a usual single-sample or two-sample procedure is used.

Selecting the best from among k normal populations is one of the most important ranking problems and has wide practical application. The problem has been formulated as follows. Let X_{ij} be independent normal variables taken from populations π_i with means μ_i and variances σ_i^2 ($i=1, \dots, k; j=1, \dots, N_i$). The population with the largest mean is called the 'best' population. The values of μ_i are, however, unknown to an experimenter. Let the ranked μ -values be denoted by:

$$\mu_{[1]} \leq \mu_{[2]} \leq \dots \leq \mu_{[k]}$$

and the differences between any pair of them be denoted by:

$$\delta_{ij} = \mu_{[i]} - \mu_{[j]} \quad (i \geq j; i, j=1, \dots, k).$$

The experimenter is interested in determining the smallest sample sizes N_i ($i=1, \dots, k$) such that

$$(1) \quad \text{Prob} \{ \text{Selecting the best population} \} \geq P^* \text{ whenever } \delta_{k, k-1} \geq \delta^*,$$

where δ^* and P^* ($1/k < P^* < 1$) are positive constants preassigned by him.

Three notable procedures have been proposed:

- (a) Bechhofer (1954) – Single-sample procedure for the case of known variances;

* W.K. Chiu is lecturer in Statistics, University of Hong Kong, Hong Kong, and visiting lecturer in Mathematics, University of Bath, Bath, England.

- (b) Bechhofer, Dunnett & Sobel (1954) – Two-sample procedure for the case of known variance ratios; and
- (c) Ofosu (1973) – Two-sample procedure for the case of unknown variances.

From these papers it appears that although much attention has been given to the knowledge that the experimenter possesses of the variances, an important question about the means:

- (2) What happens if $\delta_{k,k-1} < \delta^*$?

has not been answered. In practice, the experimenter may not know that the condition $\delta_{k,k-1} \geq \delta^*$, as required by (1), is satisfied. For example, in the case of medical trials in which the experimenter wishes to select one from several new medicines for curing a certain disease, he probably does not know whether the mean effect of the best medicine exceeds that of the second best by at least δ^* before the experiment is performed. Under such circumstances, it is desirable to provide an answer to question (2). For this purpose, we may introduce a slightly different goal for the ranking problem in question.

DEFINITION.

Given k normal populations π_i with unknown means μ_i ($i=1, \dots, k$), and a constant $\delta^* > 0$, any selection of a population π_j which satisfies the condition

$$\mu_{[k]} - \mu_j < \delta^*,$$

where $\mu_{[k]}$ denotes the largest population mean, is called a ' δ^* -correct selection'.

This definition implies that the preassigned constant δ^* is the smallest amount of difference in the population means that the experimenter wishes to detect. Since the selection of the best population is δ^* -correct, the present goal can be regarded as a generalisation of the original goal, and these two goals coincide when $\delta_{k,k-1} \geq \delta^*$.

Before presenting the main result of the present article we first note that the procedures (a), (b), and (c) cited above all satisfy the following monotonicity condition:

- (a) Prob $\{X_{(i)} < X_{(k)} \text{ (} i=1, \dots, k-t \text{)}\}$ is an

increasing function of $\delta_{i+1,j}$ ($i=1, \dots, k-1$).

We may now state and prove the following

Theorem

For any sampling procedure which selects the population with the largest

observed sample mean and for which condition (A) holds, we have that (1) implies:

(3) $\text{Prob } \{\delta^*\text{-correct selection}\} \geq P^*$.

Conversely, (3) implies (1).

Proof:

Let $\bar{X}_{(i)}$ denote the sample mean for the population having the mean $\mu_{[i]}$ ($i=1, \dots, k$), and let $t(1 \leq t \leq k)$ be the unique integer such that

(4) $\delta_{k,k-t+1} < \delta^*$ and $\delta_{k,k-t} \geq \delta^*$,

where it is understood that $\delta_{k0} = \infty$. The theorem is true if $t=k$ because then any selection is δ^* -correct, and inequality (3) holds for any P^* . For any $t < k$, we have:

$$\begin{aligned} & \text{Prob } \{\delta^*\text{-correct selection}\} \\ &= \sum_{\alpha=k-t+1}^k \text{Prob } \{\bar{X}_{(i)} < \bar{X}_{(\alpha)} \ (i=1, \dots, \alpha-1, \alpha+1, \dots, k) | (4)\} \\ &\geq \text{Prob } \{\bar{X}_{(i)} < \bar{X}_{(k)} \ (i=1, \dots, k-t) | (4)\} \\ &\geq \text{Prob } \{\bar{X}_{(i)} < \bar{X}_{(k)} \ (i=1, \dots, k-t) | \delta_{k,k-t} = \delta^*, \delta_{k-t,1} = 0\} \end{aligned}$$

This last inequality is established because of condition (A). The probability is further equal to

$$(5) \quad \text{Prob } \{\bar{X}_{(i)} < \bar{X}_{(k)} \ (i=1, \dots, k-t) | \delta_{k,k-t} = \delta^*, \delta_{k-1,k-t} = \delta_{k-t,1} = 0\}$$

because these two probabilities have the same integral expression which does not involve the populations associated with $\mu_{[k-t+1]}, \dots, \mu_{[k-1]}$.

Further, we have the following inequalities:

$$(5) \geq \text{Prob } \{\bar{X}_{(i)} < \bar{X}_{(k)} \ (i=1, \dots, k-1) | \delta_{k,k-1} = \delta^*, \delta_{k-1,1} = 0\} \geq P^*,$$

because of (1).

The converse of the theorem is obviously true, for it corresponds to the case of $t=1$ in (4). This completes the proof.

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